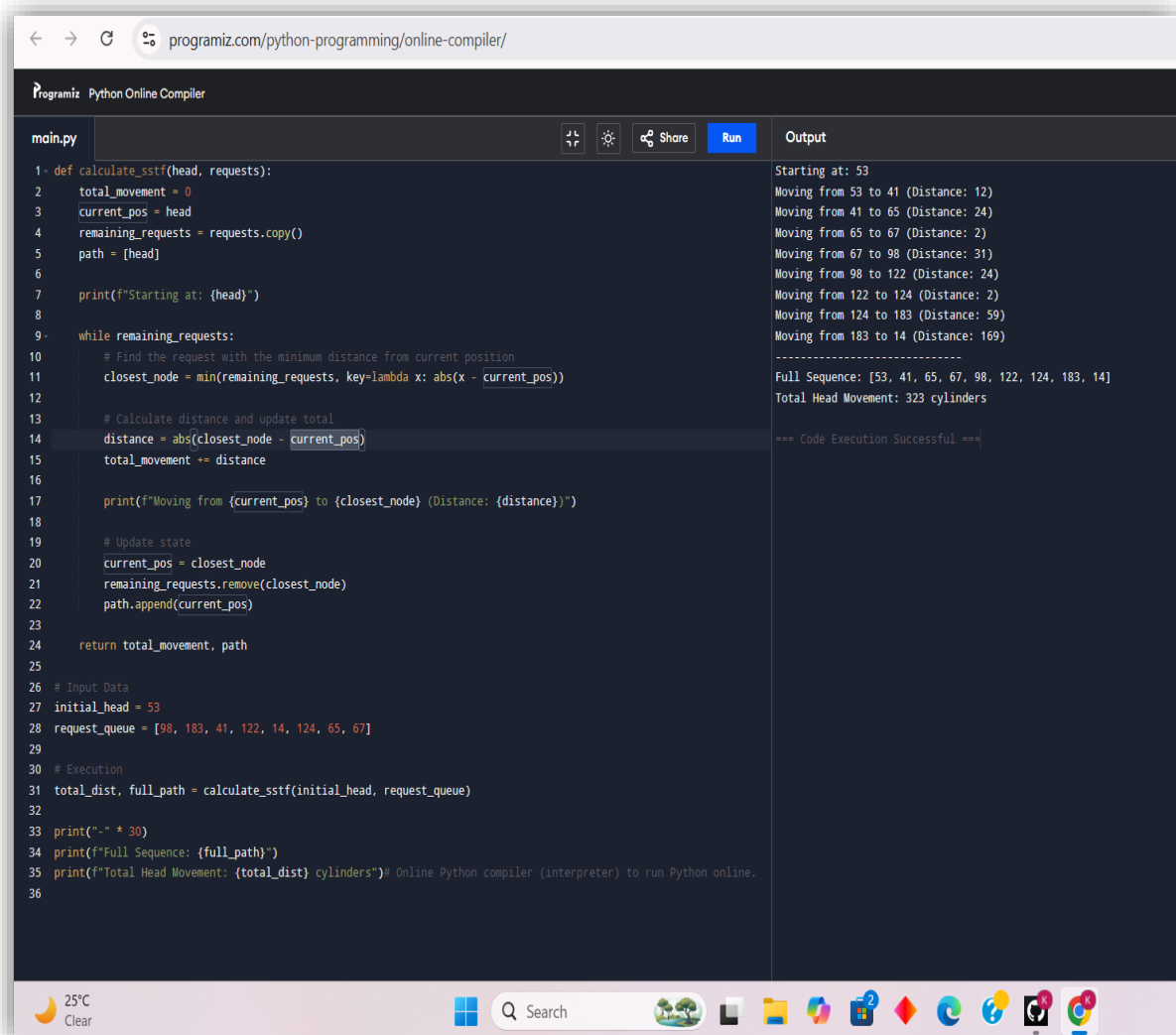


Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.



The screenshot shows a web browser window with the URL `programiz.com/python-programming/online-compiler/`. The page displays a Python Online Compiler interface. The code editor on the left contains a Python program that implements the SSTF (Shortest Seek Time First) disk scheduling algorithm. The program starts with an initial head position of 53 and a list of requests: [98, 183, 41, 122, 14, 124, 65, 67]. It calculates the total head movement by always selecting the request closest to the current head position. The output on the right shows the sequence of moves and the total head movement of 323 cylinders.

```
main.py
1- def calculate_sstf(head, requests):
2     total_movement = 0
3     current_pos = head
4     remaining_requests = requests.copy()
5     path = [head]
6
7     print(f"Starting at: {head}")
8
9     while remaining_requests:
10        # Find the request with the minimum distance from current position
11        closest_node = min(remaining_requests, key=lambda x: abs(x - current_pos))
12
13        # Calculate distance and update total
14        distance = abs(closest_node - current_pos)
15        total_movement += distance
16
17        print(f"Moving from {current_pos} to {closest_node} (Distance: {distance})")
18
19        # Update state
20        current_pos = closest_node
21        remaining_requests.remove(closest_node)
22        path.append(current_pos)
23
24    return total_movement, path
25
26 # Input Data
27 initial_head = 53
28 request_queue = [98, 183, 41, 122, 14, 124, 65, 67]
29
30 # Execution
31 total_dist, full_path = calculate_sstf(initial_head, request_queue)
32
33 print("-" * 30)
34 print(f"Full Sequence: {full_path}")
35 print(f"Total Head Movement: {total_dist} cylinders")# Online Python compiler (interpreter) to run Python online.
36
```

Output

```
Starting at: 53
Moving from 53 to 41 (Distance: 12)
Moving from 41 to 65 (Distance: 24)
Moving from 65 to 67 (Distance: 2)
Moving from 67 to 98 (Distance: 31)
Moving from 98 to 122 (Distance: 24)
Moving from 122 to 124 (Distance: 2)
Moving from 124 to 183 (Distance: 59)
Moving from 183 to 14 (Distance: 169)
-----
Full Sequence: [53, 41, 65, 67, 98, 122, 124, 183, 14]
Total Head Movement: 323 cylinders

=== Code Execution Successful ===
```